

Agentic System Design Proposal

I. Executive Summary

This proposal addresses the digital forensics domain; it describes an automated agentic software system (for brevity, the “system”) for the identification, processing, archiving, and auditing of user-generated social media digital content files. This proposal concludes that such an approach provides significant potential for more efficient user content moderation and overall labour and cost savings compared to reliance on conventional human moderators.

II. Design Acceptance Criteria

The acceptance criteria for a fully formed content moderation system revolves around three key functionality requirements:

A. Automated Actionable Content Detection

The system must be capable of identifying potentially actionable user content to an acceptable degree of accuracy without any human interaction, meaning, user-generated content that must be removed from the platform in order to comply with

the platform's terms of service and/or any other regulatory requirements (Bandara et al, 2025; Hase, 2024).

B. Automated Actionable Content Processing

The system must efficiently process content in a manner that automatically acts on, and exposes, its results to human oversight (Bandara et al, 2025).

C. Audit Archiving

The system must archive the results of its actions for future auditing, both for company-internal purposes, and for legal review by regulators and other third parties, as necessary (Bandara et al, 2025; Hase, 2024; Rondon et al, 2025).

III. Solution Architectural Structure

At the highest level, the proposed system may be described as a “pod,” to use a colloquial term of art, of interlinked software agents that act independently in terms of their “single responsibility” or individualised tasks (Bandara et al, 2025; Usman et al, 2025), but in coordination with each other to accomplish the system's ultimate objective of identifying, classifying, and removing objectionable user-generated content from the social media platform it monitors (Usman et al, 2025).

Furthermore, the system adopts a Belief-Desire-Intention agent model in which each agent maintains operational beliefs (e.g., metadata, hash matches),

goal-oriented desires (e.g., compliance verification), and executable intentions (e.g., archival and reporting functionality) (De Silva, Meneguzzi and Logan, 2020).

This design pattern was selected to align with industry best practices, as it provides the benefits of rapid development, more efficient resource consumption, reduced need for human labour, and enhanced flexibility to adapt the system to changing regulatory standards (Rondon et al, 2025; Sakurada, Leitão and De, 2022). While a monolithic architecture may reduce processing overhead, those reduced costs are outweighed by markedly improved modularity, scalability, and fault isolation in a complex, distributed, forensic operational environment (Souza, 2025).

For maximum clarity, visual overviews of the system architecture and processing workflow are presented in Appendices A through C. These agents and related components are additionally divided into two broad categories:

A. Core Agents

These agents address the first and second design acceptance criteria.

1. Monitoring Agent

This agent observes platform event streams to identify relevant content and to generate content references for consumption by other agents.

2. Archiving and Retrieval Agents

These agents work in tandem to consume content references generated by

the monitoring agent, both to archive and to retrieve content files as needed for use by other agents and/or by human reviewers.

3. Forensic Processing Agent

This agent is responsible for preserving data integrity and extracting forensic artefacts. It generates cryptographic hashes, extracts embedded metadata, and prepares structured forensic information for further evaluation by reviewers.

4. Compliance Evaluation Agent

This agent applies policy rules and compliance criteria to the processed data. Based on these checks, it assigns a classification result and risk level to the content for consumption by both agentic and human reviewers.

5. Reporting Agent

This agent generates compliance reports and notifications. It accesses archived evidence to communicate system actions to human moderators.

B. Data Stores and Services

These system data storage and related components address the second and third design acceptance criteria: automated content processing and audit archiving. In accordance with best practices, all data is structured and stored in an agent-first

format (Liu, 2024). For brevity, descriptions of these components are provided in Appendices B and C.

IV. Prototype Technical Implementation

For cost savings, the prototype will be built from open source components whenever possible. The LLM platforms used to categorise content will be outsourced to major industry providers, such as Google Vision AI, since they are too expensive to build in-house (Google, 2026).

In short, the prototype will be implemented in Docker containers for portability and scalability; it will use Python for code development, the FastAPI platform for agentic communication, the hashlib Python library for SHA-256 hashing, RabbitMQ for asynchronous messaging between components, and SQLite for storage (Docker, 2026; FastAPI, 2026; Python, 2026; RabbitMQ, 2026; SQLite, 2026).

V. Potential Drawbacks and Limitations

The potential drawbacks of this design pattern are well-known to the current industry state of the art; in short, they revolve around increased system complexity leading to suboptimal performance (Bhatti and Mughal, 2026).

A. Increased System Complexity

The need to build out at least one dozen software agents, data storage components, and inter-agent messaging components significantly increases overall

solution complexity in comparison to a single-source or monolithic solution (Bhatti and Mughal, 2026; Cihon, 2024; Passi, 2025).

B. Increased System Maintenance

Correspondingly, the wide variety of system agents and components leads to increased system maintenance requirements in terms of updating and refining each agent's source code as the libraries and related resources for such code are updated (Cihon, 2024; Passi, 2025).

C. Rapidly Changing Content Subject Matters

User-generated content constantly changes; new subject matters that fall inside compliance standards for prohibited content may arise faster than the system will be able to identify them (Shrivastav et al, 2026).

D. Human Oversight Remains a Necessity

The risk of false positive and false negative results is expected to remain non-zero, meaning that regulatory and compliance audits will require human review and analysis of system actions and outcomes (Cihon, 2024; Passi, 2025; Zhu et al, 2025). Human-in-the-loop review is therefore essential both for operational validation and to mitigate risks of bias and overenforcement (Cihon, 2024; Passi, 2025; Zhu et al, 2025).

VI. Conclusion and Recommendation

This proposal concludes that the expected benefits of an agentic system, moderated by human oversight, outweigh known potential drawbacks and should be implemented.

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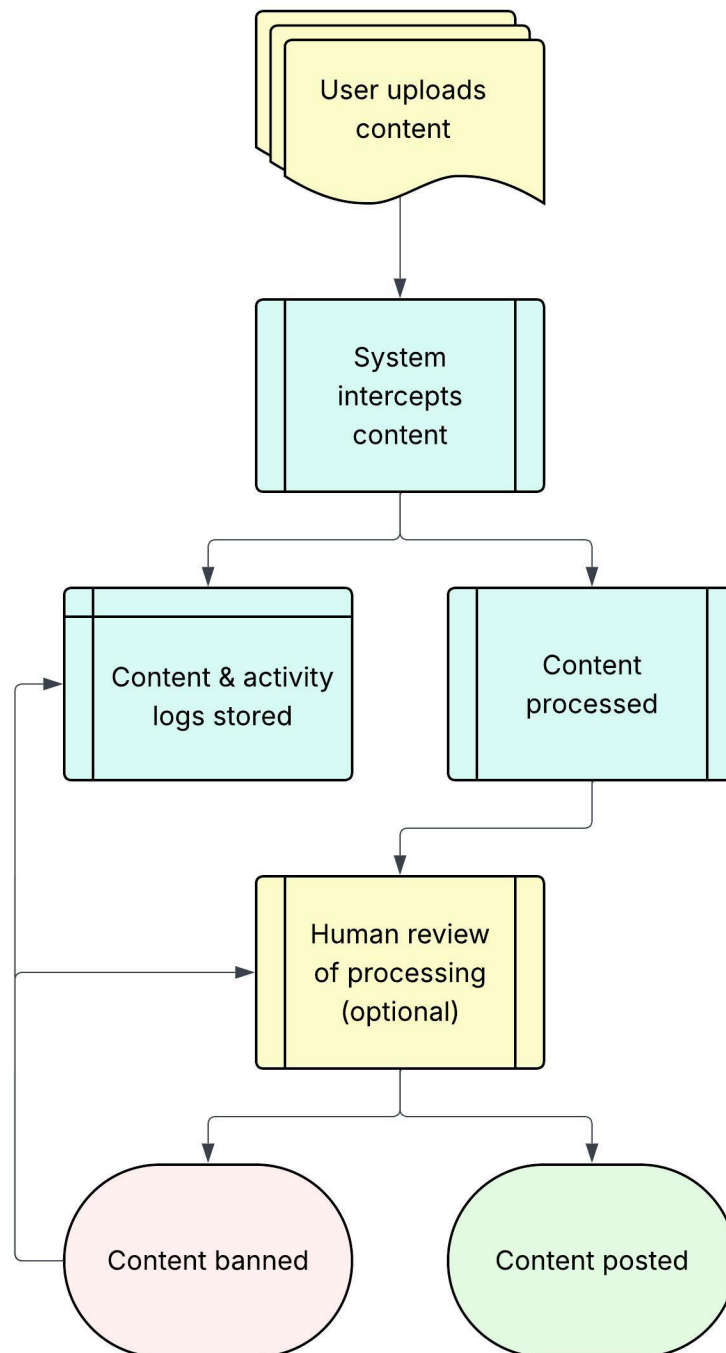
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Appendix A: Human v. Agent Actions



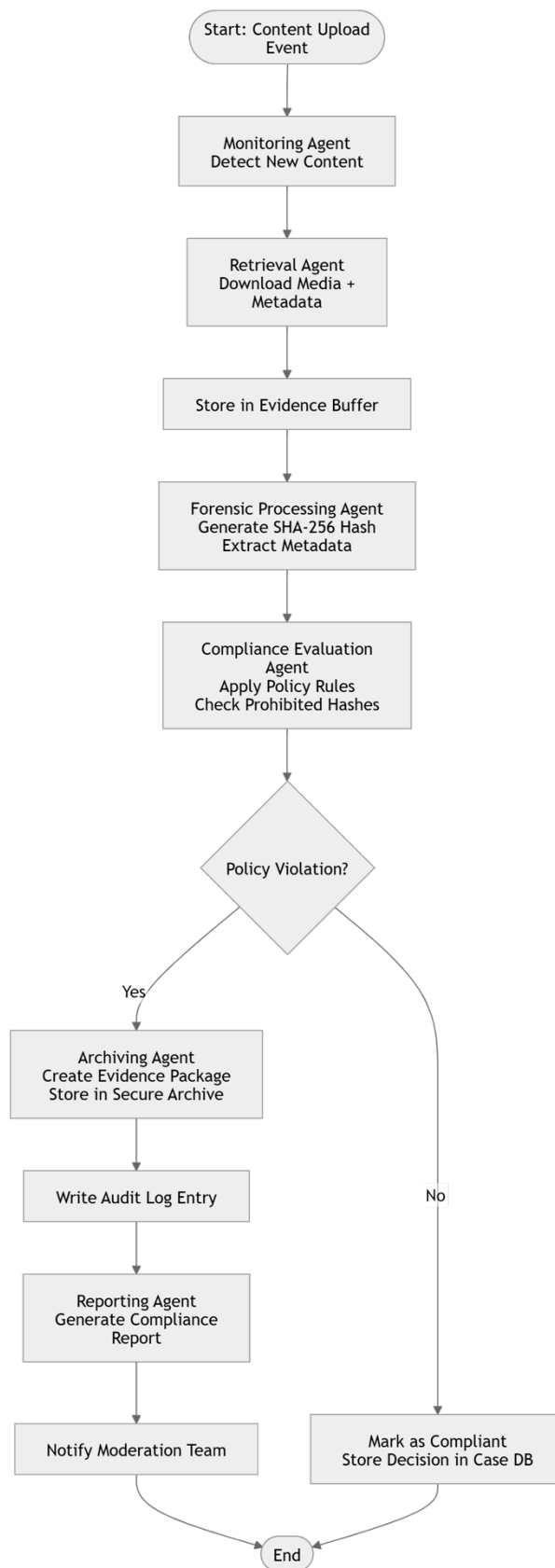
Key

Yellow: human-directed actions.

Blue: automated agent-directed actions.

Red and green: system processing outcomes.

Appendix B: System Agents & Components Workflow



Appendix C: System Agents & Components Architecture

